

Project Documentation

Data-Driven Testing Lab using Python

Objective

Develop a Python application that performs data-driven testing by reading test cases from a CSV file, comparing expected and actual results, and generating a test report automatically.

Project Scope

The project demonstrates the concept of data-driven testing by separating test data from the program logic. Test cases are stored in a CSV file, processed automatically, and summarized in a report.

Tools & Technologies

Python 3.x, CSV Module, Visual Studio Code, Git, GitHub.

Project Structure

Source_code/data_driven_testing.py

Source_code/test_data.csv

README.md

Requirements.txt

Output/test_report.txt

Screenshots/

Documentation/

Working Principle

1. Read test cases from the CSV file.
2. Compare expected and actual results.
3. Determine PASS or FAIL status.
4. Display the results in the terminal.
5. Generate a test_report.txt file.

Features

Reads test cases from a CSV file, compares results automatically, displays PASS/FAIL status, and generates a detailed test report.

Expected Output

The application displays the execution status of each test case, followed by a summary showing the total passed and failed tests. A test_report.txt file is also generated.

Future Enhancements

Support Excel input files, graphical reports, database-driven testing, automated email reporting, and integration with testing frameworks such as pytest.

Conclusion

This project provides a simple implementation of data-driven testing using Python. It demonstrates how external test data can be used to automate test execution and reporting efficiently.